

- 7 (a) In an ideal gas equation of state, the product of the pressure P and volume V of an ideal gas is directly proportional to the thermodynamic temperature T , $pV = nRT$.

At volume $2.0 \times 10^{-3} \text{ m}^3$, the pressure of the air was $1.0 \times 10^5 \text{ Pa}$ and hence $pV = 200 \text{ J}$.

At volume $8.0 \times 10^{-4} \text{ m}^3$, the pressure of the air was $4.0 \times 10^5 \text{ Pa}$ and hence $pV = 320 \text{ J}$.

The product pV has increased from 200 J to 320 J, hence the temperature has increased.

**The ideal gas equation should be used in this part of the question instead of the first law of thermodynamics.*

- (b) By taking the logarithm of both sides of the equation,

$$\lg p + \gamma \lg V = \lg c$$

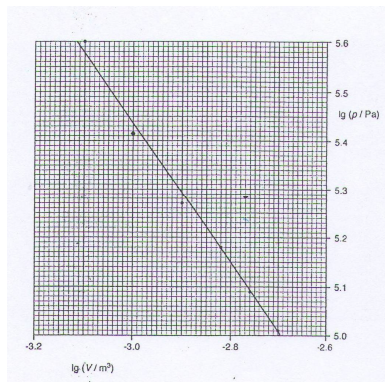
$$\lg p = -\gamma \lg V + \lg c$$

If the relationship holds, then this graph will be a straight line with the equation in the form: $y = mx + c$ where $m = -\gamma$ and y-intercept $\lg c$.

- (c) (i) $V = 1.00 \times 10^{-3} \text{ m}^3$, the pressure is $p = 2.6 \times 10^5 \text{ Pa}$, $\lg(p / \text{Pa}) = 5.41$

- (ii) 1. $V = 1.00 \times 10^{-3} \text{ m}^3$, $\lg(V / \text{m}^3) = -3.0$.

2.



**The line of best fit must have a balance of points on both sides of the best fit line.*

$$(iii) -\gamma = \frac{(5.0 - 5.6)}{(-2.69 - (-3.12))} = -1.4$$

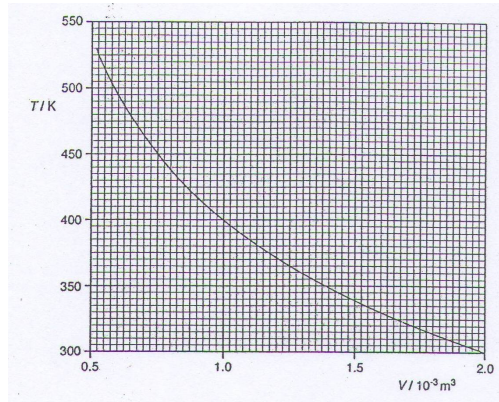
$$5.0 = -1.4(-2.69) + \lg c$$

$$c = 17$$

$$\gamma = 1.4, c = 17 \text{ kg m}^2 \text{ s}^{-2}$$

(d) For a compression ratio of 3.85, the final volume of the air is

$$\frac{2.00 \times 10^{-3}}{3.85} = 5.19 \times 10^{-4} \text{ m}^3$$



Temperature = 530 K

(e) The high pressure causes the gas to expand rapidly under adiabatic conditions, since there is no time for heat exchange between the gas and its surroundings. Applying the first law of thermodynamics, the decrease in internal energy is equal to the work done by the gas in expanding. From Fig. 7.4, the temperature decreases when the volume of the gas increases.

**The short time duration for the expansion of gas resulting in very little time for any heat exchanges with the surroundings was omitted in many explanations.*